

How Has NLP Research Changed?

A Comparative Corpus Analysis of Scientific Abstracts

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Abstract

Scientific abstracts show how researchers frame problems and key findings. This study analyzes how the framing of NLP research has changed based on a corpus of arXiv abstracts (1991-2021), comparing representation methods (TF-IDF, SBERT and LDA) and evaluating temporal shifts by keyword and semantic similarity measures. Centroid trajectories are visualised through PCA and UMAP projections. All four Axis 1 methods reflect a single dominant breakpoint at the 2011-2015 to 2016-2021 transition, which is consistent with the rise of transformer-based pre-training. The Axis 2 keyword shift confirms the same transition and cosine distance results uncover a layered pattern beneath this consensus: lexical change peaks first, semantic change later, and topical change last. NLP’s paradigm shift propagated upward through representational layers instead of occurring simultaneously.

GithubLink:<https://github.com/guruanand20/nlp-corpus-analysis>

1 Introduction

Over the past three decades, Natural Language Processing (NLP) has undergone major paradigm shifts, from rule-based and statistical approaches to data-driven machine learning and transformer-based models recently. These transitions are well documented in terms of performance, motivating a data-driven exploration of how research focus evolves over time.

Scientific abstracts provide a concise and standardised representation of research output, making them suitable for tracking such changes at scale. Analysing lexical and semantic patterns in abstracts could offer a perspective on the evolution of research focus and discourse.

In terms of shifts in vocabulary, for example, from grammar and parsing to neural and pre-trained, indicating not only new techniques but

Period	Count	Years	Era
1991–1995	448	1994–1995	Statistical
1996–2000	659	1996–2000	Statistical
2001–2005	279	2001–2005	Machine Learning
2006–2010	364	2006–2010	Machine Learning
2011–2015	2066	2011–2015	Deep Learning
2016–2021	28405	2016–2021	Transformer

Table 1: Statistics of the filtered NLP corpus.

changing conceptualisations of NLP. Identifying when these shifts occur and whether different analytical methods capture them consistently could provide insight into the dynamics of scientific change.

This project addresses the question of how the focus and vocabulary of NLP research have evolved over time and how methodological choices influence the insights obtained.

The source corpus was the publicly available `gfiessore/arxiv-abstracts-2021` dataset, which contains arXiv metadata and abstracts up to the end of 2021. From this larger dataset, the arXiv subject labels `cs.CL` and `cmp-lg` were used to filter papers related to NLP. In order to get cleaner data, publication years were extracted from arXiv identifiers, and records without reliable year information were removed. The final NLP corpus contains 32,221 abstracts covering 1994-2021.

For historical comparison, the corpus was divided into six period bins: 1991-1995, 1996-2000, 2001-2005, 2006-2010, 2011-2015, and 2016-2021. Table 1 shows the number of abstracts and actual year range in each period. Although the first bin is labeled 1991-1995, the earliest records after cleaning begin in 1994. As shown in Figure 1, the 2016-2021 period accounts for approximately 88.16% of all abstracts, which shows the corpus is strongly imbalanced. Therefore, results involving this period should be interpreted carefully because it contains far more papers than earlier periods.

A reliable method for tracking change in NLP research must meet four requirements. First, it must

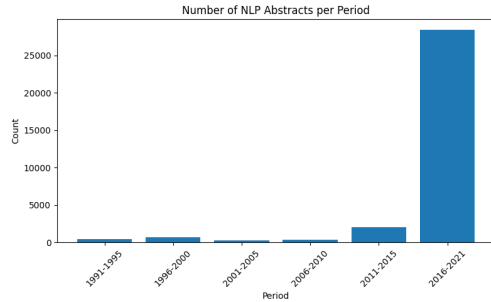


Figure 1: Number of NLP abstracts per period.

be interpretable: the output should point directly to specific words or phrases, not just numerical scores that require further explanation. Second, it must account for corpus imbalance: because the 2016–2021 period contains far more abstracts than earlier periods, any comparison must normalise for period size to avoid mistaking quantity for significance. Third, it must capture both directions of change: identifying not only which terms became prominent but also which terms disappeared from the research vocabulary. Fourth, it must be transparent enough for readers to verify the findings without specialist tools. These four requirements motivate the two analytic axes in this study. Axis 1 examines how the choice of text representation of TF-IDF versus semantic embeddings shapes what patterns become visible. Axis 2 examines how the choice of comparison method, keyword shift analysis versus cosine distance determines what kind of change can be detected and explained.

The vocabulary difference when compared between early and recent NLP abstracts shows how the field transformed over time. A 1994 paper on tree adjoining grammars focuses on derivation trees, syntactic analysis, linear indexed grammars and parsing algorithms, which showed NLP as a problem of linguistic structure and computational grammar. The research question is framed in terms of correctness of linguistic structure rather than measurable accuracy on benchmark datasets.

On the other hand, a 2018 paper about Universal Language Model Fine-tuning (ULMFiT) talks about inductive transfer learning, pre-trained models, fine-tuning and state of the art performance on benchmark datasets. The research question is framed with respect to generalization across tasks and reduction in data requirements, which were absent in the 1994 abstract.

Three differences can be noted from the above comparison. First, the interest of analysis transformed from grammatical structures to neural

model parameters. Second, the evaluation criteria shifted from linguistic correctness to measurable accuracy. Third, the framing of contribution shifted from theoretical reformulation to practical transfer and generalization.

2 Methods

2.1 Data Preparation and Preprocessing

The dataset was filtered from 1,999,486 abstracts to retain papers categorised under `cs.CL` and `cmp-lg`, resulting in 32,221 abstracts from 1994 to 2021, with the inclusion of `cmp-lg`, the predecessor of `cs.CL`. It ensures coverage of early NLP research and avoids discontinuities in the dataset. Publication years were extracted from both legacy and modern arXiv identifier formats, and documents were grouped into six five-year periods (1991-1995 to 2016-2021) to ensure comparability across time, with 2021 merged into the final bin due to limited sample size. Each period was further associated with a corresponding NLP technical paradigm (Era) for interpretation.

Text preprocessing involved lowercasing, removal of non-alphabetic characters, stopwords filtering, and lemmatisation to reduce noise and improve consistency across documents.

2.2 TF-IDF Representations

TF-IDF Unigram was adopted as a baseline representation due to its minimal assumptions and direct interpretability, enabling transparent comparison of lexical importance across periods. Representations were generated using `min_df=5` and `max_df=0.85` to remove noise and overly common terms, with `sublinear_tf=True` to reduce the dominance of high-frequency tokens, resulting in a vocabulary of approximately 15,000 terms. Given the substantial imbalance in document counts across periods, term importance was aggregated by the mean rather than the sum. We hypothesise that TF-IDF Unigram will capture broad lexical shifts between paradigms, for instance, the decline of statistical terminology and the rise of neural vocabulary. However, it tends to treat semantically related terms with different surface forms as unrelated, thereby limiting its ability to capture deeper connections in meaning.

TF-IDF Bigram extends the baseline by capturing compound expressions such as language model and neural network, which are not preserved in unigram representations. Using the same param-

eter settings, the bigram vocabulary contains approximately 18000 terms. For each period, top-ranked terms were extracted to identify emerging and declining vocabulary. For the vocabulary shift comparison, periods were divided into early (1991-1995, 1996-2000, 2001-2005) and late (2011-2015, 2016-2021). The 2006-2010 period here was excluded from this binary comparison as a transitional interval whose mixed vocabulary profile would obscure the contrast between the two paradigms. We hypothesise that TF-IDF Bigram will identify paradigm-specific compound terms, such as pre-trained model and attention mechanism that unigram representations fail to capture, providing a more precise characterisation of how NLP concepts are expressed across periods while remaining interpretable.

2.3 SBERT and LDA Representations

Based on the TF-IDF baseline, SBERT and LDA captures patterns that lexical overlap may miss. TF-IDF represents abstracts through word and phrase frequency, while SBERT focuses on sentence meaning and LDA focuses on topic structure. These two methods provided another view of the same historical change.

SBERT was implemented with the `all-MiniLM-L6-v2` Sentence-Transformers model, which maps each abstract into a dense vector with 384 dimensions (Sentence Transformers, 2024). The model produced an embedding matrix of shape $(32221, 384)$ with the full cleaned corpus. Additionally, Sentence-BERT places texts with similar meanings close together in vector space (Reimers and Gurevych, 2019). Compared with TF-IDF, SBERT can capture related ideas even when they are written with different words. The hypothesis is that SBERT will reveal semantic drift that is partly hidden from TF-IDF.

LDA was used to model topic structure at the corpus level (Blei et al., 2003). After token filtering, 32,217 abstracts remained, with a dictionary of 12,860 terms. Each abstract was represented as a topic proportion vector, where each dimension gives the probability weight of one latent topic.

In addition, candidate values of $K = 8$, $K = 10$, and $K = 15$ were compared using topic coherence, with scores of 0.5101, 0.5082, and 0.5127 respectively. As the differences were small, coherence was not the only standard to select appropriate K . The top words of the topics were also checked to interpret better. $K = 15$ was selected because it

had the highest coherence and produced meaningful, separable topics such as translation, parsing, speech recognition, neural architectures, sentiment analysis, and knowledge graphs.

The hypothesis of LDA is that LDA will show changes in research themes more clearly than TF-IDF, because it groups related words into shared topics rather than treating each word separately.

2.4 Keyword shift Representations

Keyword shift analysis tracks vocabulary change by identifying which terms are most prominent in each time period and comparing those lists across periods. For each of the six five-year bins, the mean TF-IDF score is computed across all abstracts in that period. The mean is used rather than the sum to account for the strong temporal imbalance in the corpus, ensuring scores remain comparable across periods of different sizes.

The top- N terms are extracted after filtering a custom stoplist of generic academic phrases, including paper, based, used, and present, which appear consistently across all periods without reflecting genuine research content. N is set to 20. The analysis is run separately for unigrams and bigrams to capture both individual terms and compound concepts such as language model and neural network.

For the vocabulary shift comparison, periods were divided into early (1991-1995, 1996-2000, 2001-2005) and late (2011-2015, 2016-2021). The 2006-2010 period was excluded from this binary comparison as a transitional interval whose mixed vocabulary profile would obscure the contrast between the two paradigms.

Emerging vocabulary is defined as terms present in the top- N lists of both late periods but absent from all three early periods. Disappearing vocabulary is defined symmetrically. The hypothesis is that keyword shift analysis will identify the specific vocabulary items responsible for the observed change in NLP research, something that cosine distance and embedding-based methods cannot do by design. While cosine distance measures how much two periods differ as a whole, keyword shift produces directly readable evidence: a named list of terms that rose and a named list that fell.

2.5 Cosine Distance

Cosine distance analysis measures the magnitude of change between time periods by comparing their centroid vectors in the representation space. For

each time period, a centroid is computed by taking the mean of all abstract vectors within that period:

$$\mathbf{c}_p = \frac{1}{|P|} \sum_{i \in P} \mathbf{v}_i \quad (1)$$

where $\mathbf{c}_p$ is the centroid of period p , P is the set of abstracts in that period, and $\mathbf{v}_i$ is the vector representation of abstract i . The cosine distance between two adjacent period centroids is then computed as:

$$d(p_1, p_2) = 1 - \frac{\mathbf{c}_{p_1} \cdot \mathbf{c}_{p_2}}{\|\mathbf{c}_{p_1}\| \|\mathbf{c}_{p_2}\|} \quad (2)$$

Cosine distance is well suited to high-dimensional text vectors because it measures the angle between vectors rather than their absolute distance. This makes it invariant to vector magnitude - a period with more abstracts does not produce a larger centroid simply because more vectors were averaged. This property is particularly important given the strong temporal imbalance in this corpus, where the 2016-2021 period contains 88% of all abstracts.

This method differs fundamentally from keyword shift analysis in what it can reveal. Keyword shift identifies which specific terms changed and in which direction - it produces a named list of rising and falling vocabulary items. Cosine distance instead measures how much the overall representation of a period changed relative to the previous one, without identifying the specific terms responsible. The two methods are therefore complementary: keyword shift provides interpretable evidence of what changed, while cosine distance provides a quantitative signal of when and how dramatically the change occurred.

The analysis was applied to all four representations - TF-IDF unigram, TF-IDF bigram, SBERT and LDA - producing four independent distance timelines. Comparing these timelines tests whether different representation choices lead to different conclusions about the timing of NLP’s major paradigm shifts. We hypothesize that TF-IDF based distances will detect the earliest breakpoint since vocabulary change is a surface level signal that precedes deeper conceptual shifts. SBERT distances are expected to detect a later breakpoint, reflecting when the semantic meaning of research fundamentally changed rather than just its terminology. LDA distances are expected to show the most gradual change overall, since broad research

topics shift more slowly than individual words or semantic representations.

2.6 PCA and UMAP Visualization

In order to visually and quantitatively compare the four Axis 1 representations in Section 2 and Section 2.5, each representation method must be projected on a two-dimensional plane. The right strategy depends on the structure of the source matrix.

If PCA is performed on the TF-IDF matrix, the entire sparse matrix must be transformed into a dense table. However, due to memory limitations, this is not possible for the size of our data set. Therefore, we adopted a two-step approach that follows the latent semantic analysis (LSA) framework proposed by [Deerwester et al. \(1990\)](#). The first step is to directly manipulate the sparse format with `TruncatedSVD`, reducing the dimensionality of each matrix to 50 dense dimensions. The cumulative variance grows smoothly as a function of k : When $k=50$, the cumulative explained variance of the unigrams reaches 11.6%, while the cumulative explained variance of the bigrams reaches 4.5%. For high-dimensional sparse text data, the information is widely distributed in thousands of lexical dimensions ([Deerwester et al., 1990](#)), so that no turning point is expected. According to the standard practice of LSA, we set $k=50$. In the second step, PCA reduces the dimensional output 50 to two dimensions. For unigrams, the effective variance retained is $PC1 = 0.64\%$ and $PC2 = 0.56\%$, while for bigrams they are $PC1 = 0.22\%$, $PC2 = 0.21\%$, both expressed as a percentage of the initial total variance.

Due to the high density of SBERT embeddings (384 dimensions) and LDA topic distributions (15 dimensions), the standard PCA was applied directly. The total explained variance of SBERT is 10.45%, while the LDA reaches 40.22%. The significantly high LDA values reflect its inherent lower dimensions: with only 15 topic dimensions, PCA captured a significant share of the total structure in the first two components ($PC1 = 26.6\%$, $PC2 = 13.7\%$).

To assess whether the linear projection of PCA distorts the time structure, we also applied UMAP ([McInnes et al., 2018](#)), a non-linear manifold method that preserves the local neighborhood topology. For the TF-IDF matrix, the treatment is $SVD(50) \rightarrow L2\text{-normalise} \rightarrow UMAP(\text{cosine})$; for SBERT we applied UMAP (cosine) directly; for

LDA we use UMAP(euclidean) to match the simple geometric structure of the proportions of the subject. We always set `n_neighbors=30` and `min_dist=0.3`. And set `random_state=42` to ensure repeatability. Our hypothesis is that PCA will reveal clearer and more directly comparable trends among the four representations of Axis 1, while UMAP will highlight the local subdomain structure ignored by linear projection.

3 Experimental Evaluation

3.1 TF-IDF Analysis

TF-IDF scores are used as the primary metric to track vocabulary trajectories across time, where term importance reflects relative frequency weighted by inverse document frequency. To ensure comparability across periods with highly imbalanced document counts, term importance is aggregated using the mean rather than the sum. However, this metric treats each term independently, limiting its ability to capture semantic equivalence across different surface forms.

There is a clear shift in NLP research vocabulary over time. Terms associated with statistical approaches dominate earlier periods but decline sharply after 2010, while neural-related terms such as neural, attention and pretrained show a sharp increase in frequency after 2011. The term transformer appears only in the most recent period, marking a distinct transition in research focus.

Bigram representations provide additional resolution. While machine translation remains consistently important across all periods, language model becomes dominant in the final stage, reflecting the increasing prominence of large pre-trained models. Compound terms such as pre-trained model and attention mechanism are only captured at this level, highlighting the limitations of unigram representations. Analysis of top-ranked terms across periods further shows the emergence of neural terminology alongside the decline of earlier statistical vocabulary.

These findings confirm both hypotheses: TF-IDF Unigram captures the broad paradigm shift, while Bigram reveals compound terms such as language model and attention mechanism that are not visible at the unigram level. This pattern is consistent with the keyword shift analysis (Section ??), which later identifies similar neural vocabulary emerging from 2011-2015 onwards. And the results indicate a major transition between 2011-2015 and 2016-2021,

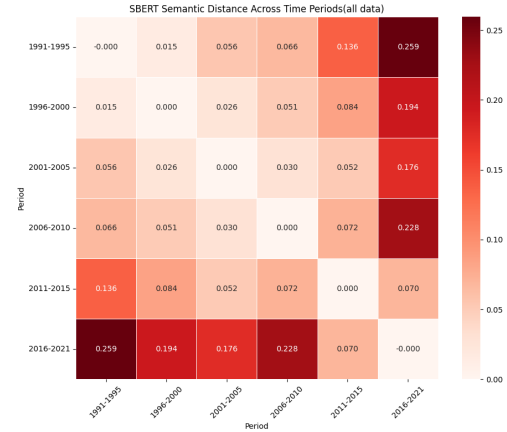


Figure 2: SBERT semantic distance across time periods.

with 2006-2010 acting as a transitional phase. The observed trends also align with established developments in NLP, including the rise of neural approaches and transformer-based models (Vaswani et al.; Devlin et al.).

TF-IDF also exhibits limitations that can affect interpretation. For instance, semantically equivalent terms such as *language model* and *language modeling* are treated as separate features, which may fragment the representation of key concepts and underestimate their true importance. Further, earlier periods with smaller sample sizes are more sensitive to outliers, which can distort term importance rankings.

3.2 SBERT and LDA Analysis

SBERT and LDA were evaluated as complements to the TF-IDF baseline. For SBERT, cosine distance between period centroids was used to measure semantic drift. As shown in Figure 2, nearby periods have smaller distances, while distant periods are more separated. The distance between 1991-1995 and 1996-2000 is only 0.015, but it rises to 0.259 between 1991-1995 and 2016-2021. This supports the hypothesis that SBERT captures gradual semantic drift rather than one sudden break. The relatively low distance between 2011-2015 and 2016-2021 also suggests that the final period continued some semantic patterns that were already present before 2016.

In terms of LDA, mean topic proportions were used to follow changes in thematic structure. The results show broader theme changes than the TF-IDF word trends, because related vocabulary is grouped into shared topics. As shown in Figure 3, Topic 8, related to parsing, grammar, and syntax, declines over time. In contrast, Topic 6, related to neural architectures, attention, and transformers,

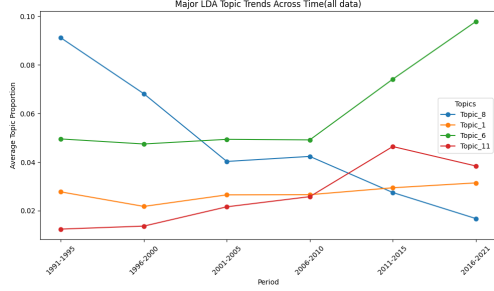


Figure 3: Major LDA topic trends across time periods.

Method	Total	Max jump	Period
TF-IDF Unigram	0.133	0.062	11-15 → 16-21
TF-IDF Bigram	0.034	0.026	11-15 → 16-21
SBERT	0.454	0.197	11-15 → 16-21
LDA ($K=15$)	0.286	0.122	11-15 → 16-21

Table 2: PCA centroid trajectory statistics (Euclidean distance in 2D projected space).

risers strongly in later periods. Topic 11, related to sentiment, social media, and bias detection, also becomes more visible after 2006-2010, while Topic 1, machine translation and multilingual NLP, remains relatively stable.

Overall, these findings are consistent with the TF-IDF result that NLP moved away from grammar and syntax vocabulary and towards neural models and transformer research. SBERT adds evidence of gradual semantic drift, while LDA shows that vocabulary change reflects broader changes in research themes. The main limitation is interpretability: SBERT shows how much periods differ, but does not indicate which terms caused the difference, and LDA topic labels require manual judgement.

3.3 PCA and UMAP Trajectories

Figure 4 shows the centroid trajectories of the PCA for the four representations. The most significant characteristic is their convergence: regardless of whether the study is represented as vocabulary frequency, sentence embedding, or subject distribution, the maximum centroid shift occurs at the same transition point (Table 2). The 2011-2015 to 2016-2021 step dominates total trajectory length in every panel, which is consistent with the emergence of pre-trained language models based on transformers after 2017 (Pramanick et al., 2023). This confirms the PCA hypothesis that a clear and consistent overall trajectory can be observed in all four representations.

The displacement magnitudes differ meaningfully. SBERT shows the largest total trajectory (0.454) and single step (0.197), consistent with the results of Section 3.1 that semantic embeddings

capture a broader and more gradual drift than lexical methods. TF-IDF Bigram’s much shorter total path (0.034 versus 0.133 for Unigrams), suggests that composite NLP terms are more stable than individual terms. PCA outputs are interpretable across four panels because they share a common Euclidean scale. Compared to UMAP, this cross-method interpretability is a significant advantage, since the magnitude of the distance across the panels in UMAP is not linearly meaningful.

The UMAP projections corroborate the PCA consensus only partially: TF-IDF bigram and LDA agree with PCA on the 2011-2015 to 2016-2021 breakpoint, but TF-IDF unigram and SBERT under UMAP both peak one period earlier (2006-2010 to 2011-2015), indicating that distributional change registered earlier in lexical and semantic space than in the linear projection.

Across the four panels of Figure 4, the three pre-transformer periods (1991-2010) cluster tightly while 2016-2021 sharply displaces to one side, which especially shows a clear plateau-then-jump structure in TF-IDF projections. Under SBERT and LDA, by contrast, the early periods trace a curved path: the early periods trace a curved arc that turns downward and rightward through 2011-2015 toward 2016-2021 under SBERT; under LDA the early-period path is similarly non-monotonic. This non-linearity, absent from TF-IDF, indicates that semantic and topical space was already reorganising during the machine-learning era (2001-2010) even as surface vocabulary remained stable, consistent with the staggered cosine distance peaks in Section 3.5.

3.4 Keyword Shift Analysis

Keyword shift analysis identifies vocabulary change at a more granular level than TF-IDF by naming the specific terms that entered and left the top-ranked lists across periods, rather than describing the general direction of change. Mean TF-IDF scores were used as the primary metric, with $N = 20$ terms per period for both unigrams and bigrams. Mean normalisation was applied to account for corpus size imbalance. The early group consisted of 1991-1995, 1996-2000, and 2001-2005; the late group consisted of 2011-2015 and 2016-2021.

The results confirmed the hypothesis. Among unigrams, the specific terms that emerged in the late periods included *neural*, *learning*, *models*, *training*, and *tasks*, none of which featured in the top-20

Research Focus Trajectory — Dimensionality Reduction Comparison
NLP Abstracts 1991–2021 | All Four Representation Methods

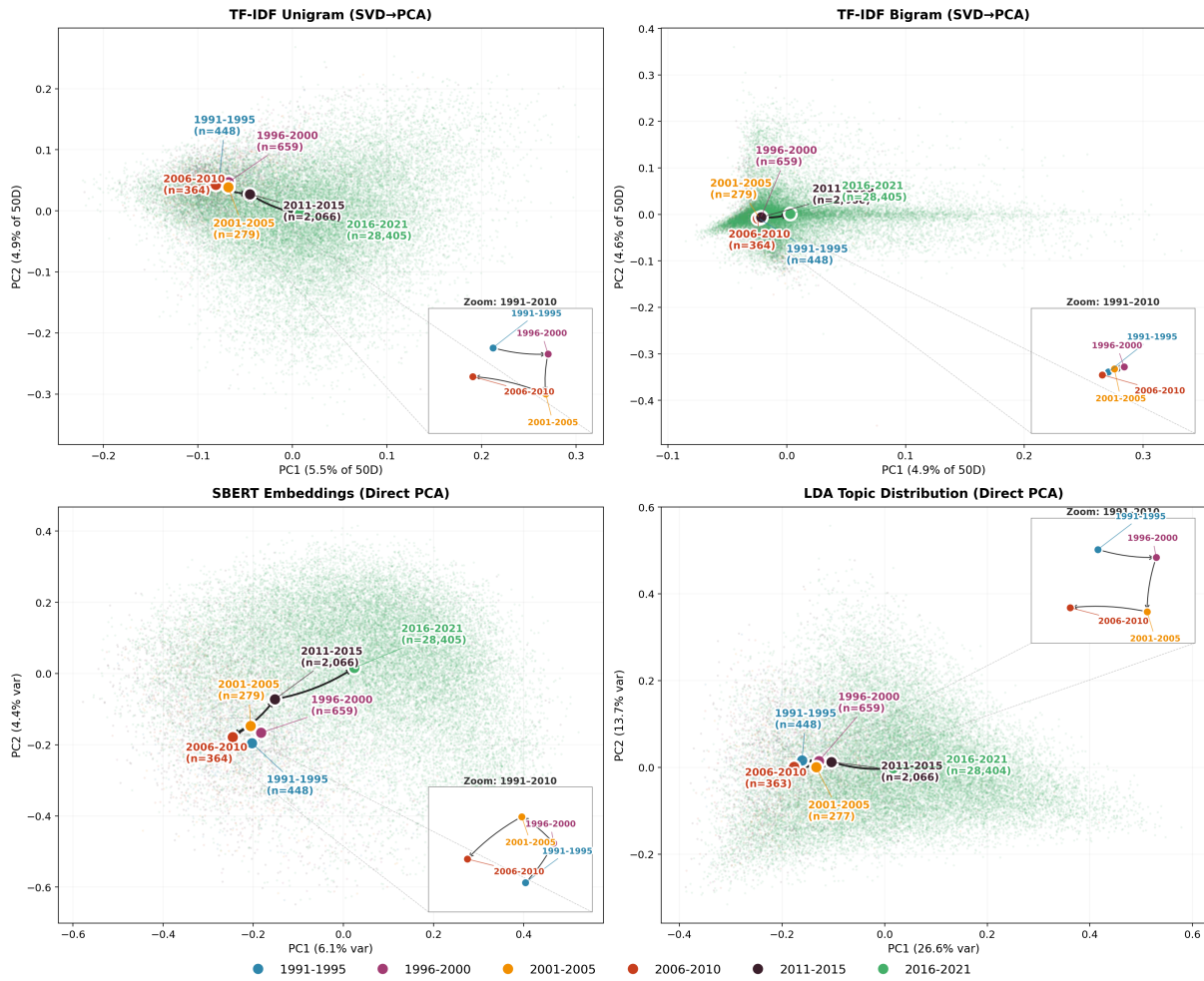


Figure 4: PCA centroid trajectories (1991–2021) for all four Axis 1 representations. Arrows connect consecutive 5-year period centroids. Zoomed insets reveal the 1991–2010 cluster, which is otherwise compressed by the scale of the 2016–2021 centroid ($n=28,405$; 88% of the corpus).

lists of any early period. The specific terms that disappeared included *grammar*, *parsing*, *unification*, *syntactic*, and *statistical*. At the bigram level, the phrase-level terms that newly entered the top-ranked lists included *deep learning*, *pre trained*, *language model*, and *word embeddings*, while *finite state*, *tree adjoining*, and *hidden markov* were absent from all late-period lists entirely. These specific vocabulary replacements align with documented developments in NLP history, including the Transformer architecture in 2017 and BERT in 2018 (Vaswani et al.; Devlin et al.).

Keyword shift analysis offers high interpretability, as each finding traces directly to named vocabulary items. This contrasts with cosine distance, which quantifies the magnitude of change but cannot identify which specific terms drove it.

Two limitations apply. First, the method cannot

distinguish genuine research shifts from changes in writing convention. Second, despite mean normalisation, the larger size of the 2016–2021 period may still cause its vocabulary to over-represent specialised subfields rather than field-wide trends. Despite these limitations, the results consistently indicate a major vocabulary shift beginning in the early 2010s, with 2006–2010 acting as a transitional phase.

3.5 Cosine Distance Analysis

The primary measure used in this analysis is cosine distance, it is defined as $1 - \cos(\theta)$ where θ is the angle between the two centroid vectors. A cosine distance value of 0 implies identical period representations, whereas a value of 1 indicates maximum divergence. This is best suited for high dimensional text vectors as it isn't dependent on

Research Focus Trajectory — UMAP Non-linear Projection NLP Abstracts 1991-2021 | All Four Representation Methods

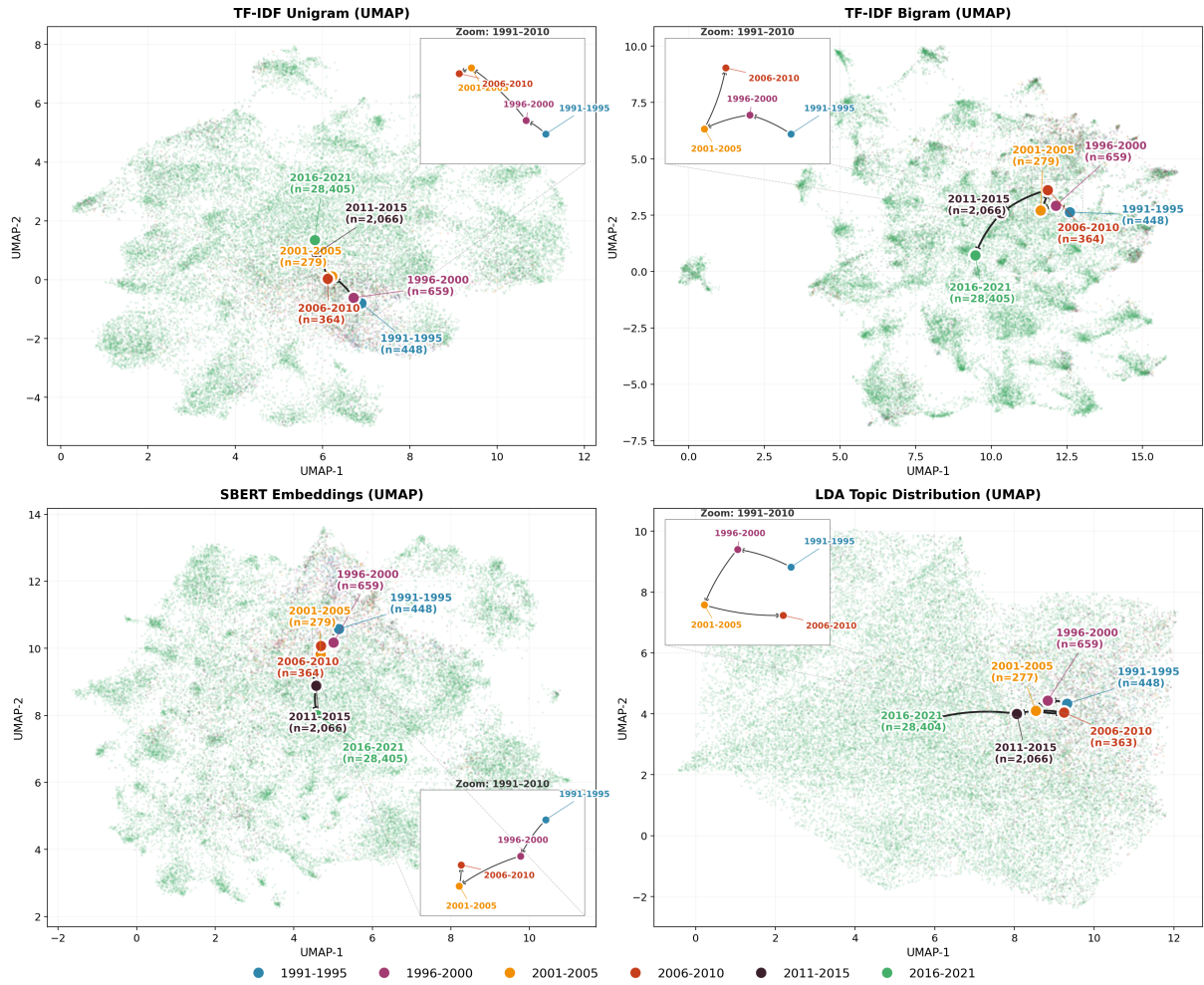


Figure 5: UMAP centroid trajectories for all four Axis 1 representations.

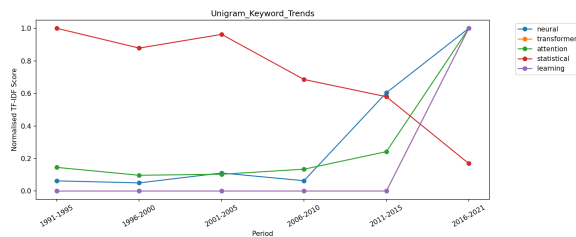


Figure 6: Normalised TF-IDF scores for selected unigram terms across periods.

vector magnitude, meaning differences in abstract count across periods do not artificially inflate the measured change. However, centroid averaging compresses the full distribution of abstracts within a period into a single representative vector, potentially masking internal heterogeneity within a period.

Cosine Distance was computed on four representations: TF-IDF unigram, TF-IDF bigram, SBERT embeddings and LDA topic distributions, all devel-

oped from the 32,217 abstract corpus after LDA token filtering. For each representation, one centroid vector was computed per period by taking the mean of all abstract vectors within that period. Distances were then calculated between every pair of adjacent periods, resulting in five transition scores across 1991 - 2021. All four representations were verified to share identical indices prior to analysis, confirming no alignment issues between the matrices.

The above table, table 3 reports the cosine distances for all period transitions across all four representations. The largest shift detected by each method is highlighted in bold in Table 3.

Figure 7 shows all four timelines side by side. Notably, bigram distances are consistently larger than unigram distances across all transitions, confirming that domain-specific phrases are more era-specific than individual words.

Transition LDA	Unigram	Bigram	SBERT
1991-1995 to 1996-2000	0.0763	0.3397	0.0152
1996-2000 to 2001-2005	0.1117	0.4381	0.0263
2001-2005 to 2006-2010	0.1295	0.5482	0.0302
2006-2010 to 2011-2015	0.1127	0.4466	0.0721
2011-2015 to 2016-2021	0.0845	0.2449	0.0698
0.0639			

Table 3: Cosine Distances between adjacent period centroids for TF-IDF unigram, TF-IDF bigram, SBERT and LDA representations

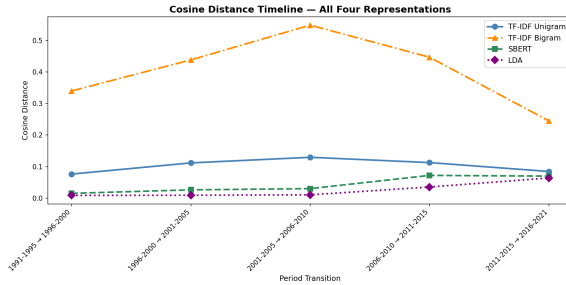


Figure 7: Cosine distance for TF-IDF unigram, TF-IDF bigram and SBERT representations across adjacent period transitions.

Three limitations affect the reliability of these results. Firstly the corpus is heavily imbalanced: the 2016 - 2021 period has around 28,405 abstracts compared to 277 in 2001 - 2005. This means the 2016-2021 centroid is computed from vastly more data, potentially underestimating the true magnitude of change entering the transformer era. Second, SBERT distances are consistently smaller than TF-IDF distances across all transitions, which could reflect genuine semantic continuity or could indicate that the pre trained SBERT model encodes a modern NLP bias. Third LDA distances are the smallest overall, suggesting broad research topics shift more slowly than vocabulary or semantic meaning, however with only 15 topics the model may be too coarse to detect finer grained thematic transitions.

3.6 Evaluation and Discussion

The most robust finding of this study is the unanimous agreement on timing. All four Axis 1 representations identify the 2011-2015 to 2016-2021 transition as the dominant breakpoint under PCA (Table 2), and the keyword shift analysis in Section 3.5 confirms the same period as the point at which neural and transformer vocabulary fully displaced statistical terminology.

Directional analysis reveals a representational divide. TF-IDF unigram (-31.9°), LDA at (-5.7°) and TF-IDF bigram (19.4°) cluster within 51.4° , while SBERT at 42.9° shows only WEAK agreement with unigram (74.8°). TF-IDF and LDA both operate on vocabulary-level structure while SBERT’s capture a different dominant variance axis. The breakpoint consensus in Table 2 nonetheless stands as the primary, more robust finding.

Axis 2 cosine distance results showed extra temporal details. TF-IDF cosine distance reaches its peak at 2001-2005 to 2006-2010, SBERT peaks at 2006-2010 to 2011-2015, and LDA peaks at 2011-2015 to 2016-2021. It suggests that NLP’s paradigm shift spreads upward through representational layers. Vocabulary first, semantic meaning second, thematic structure last. The NLP’s semantic and topical space was already reorganising during 2001-2010, even as its lexical space remained stable.

The methods differ in how directly their output can be communicated. Keyword shift analysis and TF-IDF produce named vocabulary lists any reader can verify. Cosine distance quantifies magnitude but not source. LDA requires manual topic-word inspection. SBERT trajectories require probing experiments. Within the same method, we can at least compare PCA centroid positions across panels and use UMAP to qualify structural observation. For non-specialist communication, TF-IDF and keyword shift provide the best balance. SBERT is preferred to catch hidden semantic changes where vocabulary is stable.

The 2011-2015 to 2016-2021 breakpoint matches the appearance of Transformer architecture (2017) and BERT and GPT-style pre-training (2018) (Pramanick et al., 2023; Hall et al., 2008). The earlier TF-IDF cosine peaks at 2001-2005 to 2006-2010 also map to the displacement of rule-based methods by SVMs and CRFs. (Hall et al., 2008). Taken together, these three perspectives support a single integrated reading. NLP undergo a layered reorganization that began earlier and consolidated at that boundary.

4 Conclusions

NLP research focus has changed most decisively at the 2011-2015 to 2016-2021 boundary, marking the consolidation of the transformer era. The full multi-layered cascade pattern is only visible by combining representation methods and comparison

Method	Cons.	Interp.	Align
TF-IDF Unigram	H	H	H
TF-IDF Bigram	H	H	H
SBERT	H	L	H
LDA	H	M	H
Keyword shift	H	H	H
Cosine distance	M	M	H
PCA	H	M	H
UMAP	M	L	M

Table 4: Method evaluation on three criteria (H = High, M = Medium, L = Low, Cons. = Consistency, Interp = interpretability, Align = alignment).

metrics.

TF-IDF analysis reveals how NLP research vocabulary has evolved, showing a decline in statistical terminology, a transitional phase around 2006-2010, and a rapid rise of neural and pre-training related terms after 2011.

The bigram representation refines this picture by capturing paradigm compound expressions that are not visible in unigram analysis. Both representations identify the same transition point, suggesting a robust structural change in the field.

A remaining limitation is the inability of TF-IDF to account for semantic continuity when terminology changes. Addressing this issue requires representations that operate beyond surface-level tokens, motivating the use of semantic models such as SBERT.

SBERT and LDA extend the TF-IDF findings by showing that NLP’s evolution was not only a vocabulary shift, but also a semantic and thematic change. SBERT shows gradual semantic drift across the timeline, while the close distance between 2011-2015 and 2016-2021 suggests continuity between deep learning and transformer research. LDA supports this by showing that earlier structural approaches became less dominant, while later research focused more on neural models and transformer-based methods. These results suggest that vocabulary change reflects broader changes in research meaning and topic structure. One open challenge is interpretability: SBERT distances are hard to explain directly, and LDA topic labels require manual judgement.

Keyword shift analysis revealed a sharp vocabulary transition at the 2011–2015 to 2016–2021 boundary. Terms associated with rule-based and statistical methods, including *grammar*, *parsing*, and *unification*, declined steadily and were absent from late-period top-*N* lists, while neural-era terms including *neural*, *training*, and *language model* were absent before 2011 and dominant thereafter.

The method’s key strength is specificity: it names the words that changed, not only the fact that change occurred. A remaining open challenge is determining whether a rising term signals a genuine shift in research direction or a change in writing convention, as term frequency alone cannot separate scientific transition from linguistic trend.

Cosine distance analysis showed that the NLP shift over the years depends on which representation is used to measure it. Applied among four representations, TF-IDF unigram and bigram both indicated the largest vocabulary shift from period 2001-2005 to 2006-2010, SBERT identified the shift one period later 2006-2010 to 2011-2015, and LDA identified the shift one period later 2011-2015 to 2016-2021. One open challenge remains: the corpus is heavily imbalanced, with 88% of abstracts concentrated in the 2016–2021 period. This raises the question of whether the detected shifts reflect field-wide transitions or are partly an artefact of the growing volume of abstracts in recent years inflating the stability of later period centroids.

The strongest findings came from triangulating across representations and metrics. No single method gave the complete picture. In lexical space, NLP’s pre-transformer history looks like a stable plateau ending in a sudden break. But in semantic and topical space, it appears as a continuous non-linear reorganization.

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